

Tower Building

(Coach Scheduled Event - 45 Minutes)



Description:

Teams will construct a tower using paper drinking straws and tape. The tower will be evaluated on its height, strength, and engineering knowledge. Towers will be tested using progressively heavier loads to determine the maximum load successfully supported. Teams will also complete a written assessment covering tower design, structural forces, stability, and engineering principles.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Teams Must Bring:

- Pencil, pen, or marker

Teams May Bring:

- Ruler or straightedge
- One pair of scissors per team member
- Four Function Calculator

Materials Provided at Event:

- 30 - 20 cm (8 in) paper drinking straws
- 1 meter of 2.54 cm (1-inch) masking tape
- One wooden loading block approximately 5 cm × 5 cm × 2 cm (2 in x 2 in x 3/4 in)
- One 50 g (1.8 oz) mass capable of sitting on the loading block
- One 100 g (3.5 oz) mass capable of sitting on the loading block
- One 300 g (10.6 oz) mass capable of sitting on the loading block

THE COMPETITION

Construction Phase (25 minutes):

The tower must:

- Stand freely on a flat testing surface.
- Be constructed entirely from the provided straws and tape. Teams may cut, bend, or crimp the straws and may slide them inside one another. Teams may not bring any preconstructed components or use materials not provided by the Event Supervisor.
- Support the loading block at its highest point.
- Have all structural supports resting directly on the testing surface.
- Remain self-supporting without external assistance. The team may not attach any portion of the tower to the testing surface.

The Written Assessment: Simultaneously with the device testing time, the team will take a written assessment. Teams will be called up from their written assessment to test their device and then will return to their written assessment while other teams test their device. The assessment topics may include:

- Types of towers
- Tower design principles
- Structural forces (tension, compression, shear, torsion)
- Stability and center of gravity
- Load paths and load distribution
- Failure modes
- Engineering applications of towers and tall structures

Device Testing Setup

- At the direction of the Event Supervisor, teams will place their tower on a flat, level testing surface.
- No sticky surface of the tape may contact the testing surface.

Device Testing Measurements

- Before testing, the Event Supervisor (ES) will record:
 - Tower height to the nearest millimeter - Tower height will be measured vertically from the testing surface to the highest point capable of supporting the loading block. The loading block is not included in the height measurement.
 - Number of unused, complete straws - A complete straw is one that remains uncut and unmodified.
 - If a tower is freestanding and can hold the loading block. Towers that are neither freestanding or able to hold the loading block will be tested to the best of the Event Supervisor's ability in order to give the team the experience, but will score zero for both the Height and the Load Score.

Load Testing

- Teams will place the loading block on the tower at its highest supported point.
- Teams will then test their tower using progressively heavier loads.
 - 50 g (1.8 oz) mass for 30 seconds
 - 100 g (3.5 oz) mass for 30 seconds
 - 300 g (10.6 oz) mass for 30 seconds
- A load is considered successfully supported if:
 - The tower remains standing.
 - The load remains supported.
 - The loading block remains supported.
 - No portion of the tower touches the tabletop except its intended base supports.

- Thirty seconds elapse without failure.
- Testing ends when:
 - The tower collapses.
 - The load falls.
 - The loading block falls.
 - The tower is touched by competitors after testing begins.
 - Any portion of the tower contacts the tabletop other than its intended base supports.

SCORING: Final Score (FS) = Height Score + Load Score + Assessment Score

- Height Score = $(TH \div CH) \times 40$, maximum is 40 pts
 - TH (Team Height) = Measured height of the team's tower
 - CH (Competition Height) = Height of the tallest freestanding tower able to hold the loading block
- Load Score = $(TL \div CL) \times 40$, maximum is 40 pts
 - TL (Team Load) = the maximum number of grams of the load successfully supported by the team's tower (either 50, 100 or 300)
 - CL (Competition Load) = the maximum number of grams of the load supported by any freestanding tower
- Assessment Score = $(TAS \div CAS) \times 20$, maximum is 20 pts
 - TAS (Team Assessment Score): Team written assessment score
 - CAS (Competition Assessment Score): Highest written assessment score across teams

Scoring Example

- Team's height (TH) = 12 cm. The Competition Height (CH) = 15 cm
 - Height Score = $(12 \div 15) \times 40 = 32$ points
- The team's tower (TL) held 300 g. That was the most of all the teams, so CL = 300.
 - Load Score = $(300 \div 300) \times 40 = 40$ points
- Competition Assessment Score (CAS) = 10. The team scores 6 on the assessment (TAS).
 - Assessment Score for the team = $(6 \div 10) \times 20 = 12$ points
- Final Score (FS) = $32 + 40 + 12 = 84$

Tiebreakers:

- Number of unused straws
- Select test questions

Resources:

- [Building for Hurricanes: Engineering Design Challenge](#)
- [Teach Engineering - Straw Towers to the Moon](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles